

Auditable multimodal integration of metagenomic sequencing and clinical evidence identifies causative pathogens in lower respiratory tract infection

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Abstract

Metagenomic next-generation sequencing (mNGS) detects nearly every organism in a respiratory specimen, but most of what it reports is colonizing, contaminating or reactivated rather than causative, and the physician is left to reconcile a long organism list with cultures, multiplex PCR, antigen tests, imaging, blood counts and host status. Here we present AETIA (Agentic Evidence Tracing of Infectious Aetiology), a pipeline that turns this multimodal record into a short, traceable list of causative pathogens. Modality-specific language-model agents extract structured evidence from each test; a deterministic scorer grades every mNGS organism by read burden, within-specimen dominance, control-subtracted rank, specimen alignment and same-organism hospital support, and applies pre-specified guardrails against known colonizers; a language-model safety review flags what the rules did not pick; and a final rule rescues a candidate only when PubMed literature judged against the patient’s own site, syndrome and host, or definitive direct evidence, supports it. Selection is frozen before any rationale is written, and every decision carries its rule path and source hashes. On [55] patients with suspected lower respiratory tract infection from three Taiwanese hospitals, AETIA reached a precision of [71.6%], recall of [91.4%] and F1 of [80.3%] against physician-adjudicated pathogens, compared with F1 of [14.8%] for the raw mNGS report, [50.0%] for FilmArray and galactomannan testing and [45.5%] for culture, and, in an earlier 35-patient intensive-care series, reducing the average number of organisms presented per patient from [5.8] to [2.0]. On [41] patients, the same pipeline exceeded four frontier language models prompted directly with the identical raw data by [14–15] F1 points, mainly through recall ([90.8%] versus [57–61%]). AETIA is research software whose output is a reviewed candidate list, not a diagnosis; it shows that structured, literature-grounded integration, rather than end-to-end prompting, is what makes mNGS clinically interpretable.

Introduction

Lower respiratory tract infection remains the infection most often treated without a microbiological diagnosis. In the largest prospective aetiological study of community-acquired pneumonia requiring admission, no pathogen was identified in 62% of patients despite exhaustive conventional testing¹, and in the intensive care unit, where prior antibiotic therapy reduces culture yield and the differential diagnosis includes fungi, viruses and opportunistic pathogens, the yield is lower still. Metagenomic next-generation sequencing (mNGS) of respiratory fluid or plasma was introduced to close this gap: a single assay that sequences all nucleic acid in the specimen and can, in principle, detect any bacterium, fungus, virus or parasite without a prior hypothesis^{2,3}. Its diagnostic value has been demonstrated in meningoencephalitis⁴, bloodstream infection^{5,6} and lower respiratory tract infection^{7–10}, and it is now ordered routinely for critically ill patients in many hospitals in Asia^{11,12}.

The difficulty has moved from detection to interpretation. Because mNGS is hypothesis-free, it reports everything: the upper-airway flora that every bronchoalveolar lavage carries^{13,14}, reagent and environmental contaminants¹⁵, latent herpesviruses that reactivate under critical illness, and *Candida* species that colonize the airway in almost all patients receiving mechanical ventilation. The cost of that comprehensiveness is measurable in published series. Across 511 clinical specimens, mNGS reached a sensitivity of 50.7% for infectious disease but a specificity of 85.7%¹¹; in a prospective bronchoalveolar lavage cohort it identified a definite or probable pathogen in 65% of patients with lower respiratory tract infection, as compared with 20% by culture, yet reached a specificity of 75.4% and a positive predictive value of 78.5% against a composite reference standard⁹. Separating colonization from infection is identified as the central unresolved challenge in systematic assessments of the method^{3,10}. Laboratories mitigate this with read thresholds and no-template-control subtraction^{16,17}, but a threshold cannot distinguish a *Klebsiella* with tens of thousands of reads in a patient with new bilateral consolidation and rising C-reactive protein is a pathogen, whereas a *Candida albicans* with similar reads in the same specimen is not. That judgement requires the culture and multiplex polymerase chain reaction (PCR) results, the galactomannan, the imaging, the blood counts, coexisting conditions and immunosuppression, and the timing of each test relative to the specimen, and it requires knowledge of what each organism does at that site in that kind of host. In practice it is made by an infectious diseases physician reviewing a dozen separate reports, and it is neither standardized nor recorded.

Sequencing does not retire the conventional tests, because the two fail in different directions. Culture alone yields a viable isolate and therefore phenotypic antimicrobial susceptibility, on which guideline-directed de-escalation depends^{18,19}, whereas mNGS reports nucleic acid whether or not the organism is viable and resistance inferred from sequence is not yet a substitute^{2,3}. Multiplex PCR panels return a result within an hour, against thresholds fixed at validation²⁰. Antigen assays measure something sequencing never sees: galactomannan is a mycological criterion for invasive fungal disease in its own right²¹, and antibody responses identify organisms whose nucleic acid may never be abundant in the airway. Imaging establishes that pneumonia exists at all and where it is; the guideline definition of community-acquired pneumonia requires a radiographic infiltrate, not a positive microbiological test¹⁸. Two limits are structural rather

than technical. First, a specimen reports only its own compartment, so an organism walled off in an empyema or abscess, or confined to blood or tissue, is absent from lavage fluid however deeply that fluid is sequenced. Second, detectability is a window: intermittent bacteraemia, a resolving viraemia and nucleic acid already cleared by effective therapy are each invisible to a specimen taken outside it, which is why specimen type, specimen selection and timing govern the yield of the assay^{17,22,23}. For these conditions and organisms mNGS will return false negatives at any sequencing depth, and the finding that corrects the record comes from a culture bottle, an antigen result or a chest scan. Causative diagnosis therefore rests on no single test but on the intersection of all of them, which makes it a multimodal problem before it is a sequencing problem.

Integrating that many sources, for every organism on a long report, is the task for which large language models are an obvious candidate. They encode clinical knowledge²⁴, produce credible differential diagnoses from case vignettes^{25,26} and have been tested as consultants in infectious diseases²⁷. But performance on licensing examinations and curated vignettes has not transferred to realistic clinical work: across 2,400 real patient cases, current models diagnosed significantly worse than physicians, followed neither diagnostic nor treatment guidelines and could not interpret laboratory results²⁸. The models also give no account of which test tipped the decision, cannot be audited for whether they used a finding they should not have seen, and change their answer from one run to the next²⁹. Multi-agent prompting frameworks^{30–32} and retrieval augmentation^{33,34} address parts of this, but a pipeline that will sit between a sequencing laboratory and a prescriber needs more than a better prompt. It needs decisions that are reproducible, evidence that is traceable to a source rather than to a post-hoc account of the model’s own behaviour³⁵, a guarantee that the reference standard never leaks into the reasoning, a form of leakage that is a frequent cause of inflated performance in machine-learning-based science³⁶, and a clear separation between what the model selected and how it later explained the selection, because a model’s stated reasoning need not be the reasoning that produced its answer³⁷.

Here we describe AETIA (Agentic Evidence Tracing of Infectious Aetiology), a pipeline built to those requirements and evaluated on patients with suspected lower respiratory tract infection at three medical centres in Taiwan. Single-modality language-model agents extract structured evidence, a deterministic scorer and pre-specified guardrails select organisms, case-matched PubMed literature can rescue a candidate the rules left out, and a rationale is written only after the selection is frozen. AETIA identifies causative pathogens more accurately than the raw mNGS report or any single conventional read-out, reduces the number of organisms presented to the physician, and outperforms frontier language models prompted directly with the identical de-identified data, chiefly through recall. Every decision replays byte-for-byte from frozen intermediates and carries its rule path and the hashes of the inputs that produced it. AETIA is research software whose output is a candidate list for clinical review, not a diagnosis; its contribution is to show that structured, literature-grounded integration, rather than end-to-end prompting, is what makes mNGS interpretable at the bedside.

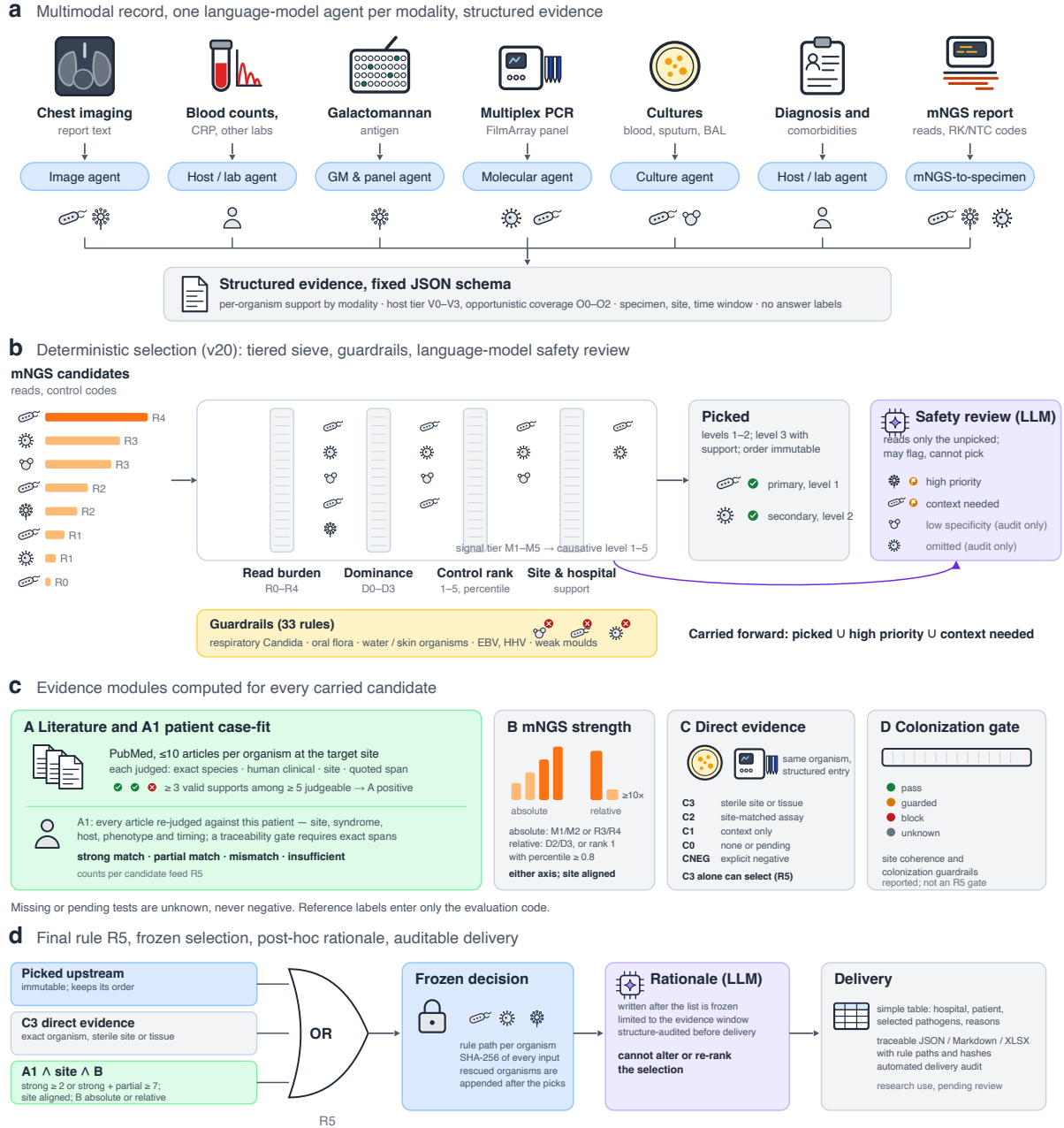


Figure 1: Overview of AETIA. **a**, One patient's record. Seven read-outs are each parsed by a dedicated language-model agent into a fixed schema; glyphs mark the organism classes (bacterium, fungus, virus, yeast) or host context an agent supplies. Agents see neither the reference labels nor each other. **b**, Deterministic selection. Organisms on the mNGS report (bars, reads; R0–R4, read tier) pass four filters—read burden, within-specimen dominance, control-subtracted rank, and site alignment with same-organism hospital support—that assign a signal tier and a causative level; 33 guardrails demote colonizers and reactivation viruses (crossed glyphs). Levels 1–2 and supported level 3 are picked in a fixed order. A language-model review reads only the unpicked organisms and can flag them (high priority, context needed) but cannot pick. **c**, Evidence for each carried candidate: A, PubMed articles judged for exact-species, site-matched human evidence and re-judged against the patient (A1 case-fit); B, mNGS strength on an absolute and a relative axis; C, direct non-mNGS evidence graded C0–C3; D, colonization gate (reported, not gating). **d**, Rule R5 selects organisms that were picked, meet C3, or meet A1, site alignment and a B axis together. The list is frozen with rule paths and input hashes before a language model writes the rationale; delivery is a summary table with a traceable JSON, Markdown and XLSX package.

Results

AETIA separates evidence extraction, rule-based selection and literature-grounded rescue

AETIA takes, for one patient, the standardized record of nine sources: admission diagnosis, underlying conditions, blood counts, other laboratory values, cultures, FilmArray results, galactomannan, imaging reports and the laboratory’s mNGS report with per-organism read counts and no-template-control codes (Fig. 1a). Each modality is read by its own language-model agent into a fixed JSON schema. The agents are single-purpose: the imaging agent sees only imaging, the culture agent only cultures, and none of them sees the answer labels. A deterministic summary then joins the per-modality outputs into organism-level support and a host-vulnerability tier.

Selection is deterministic (Fig. 1b). Every organism on the mNGS report is ranked by its control-subtracted priority and read percentile, tiered by read burden (R0–R4, decade thresholds from 1 to 10,000 reads) and by dominance within the specimen (D3 when the top organism has at least ten times the reads of the runner-up), aligned with the target site, and joined with same-organism hospital evidence. These features map to an mNGS signal tier (M1 strong to M5 background) and, together with site alignment and host status, to one of five causative levels. Pre-specified guardrails demote respiratory *Candida*, oral and upper-airway flora, waterborne Gram-negative environmental organisms, skin flora, Epstein–Barr virus and other low-actionability herpesviruses, weak mould signals and generic hospital labels. Level 1–2 organisms and level 3 organisms with a protected or supported rationale are picked, in a fixed order that no later stage may change. A language-model safety review then examines only the unpicked organisms and sorts them into high-priority, context-needed, low-specificity or omitted; the first two tiers, and only those, are carried forward.

For every carried candidate, four evidence modules are computed (Fig. 1c). Module A retrieves up to ten PubMed articles for the organism at the target site and judges each for exact-species, human, site-matched causal evidence with a quoted supporting span; module A1 re-judges every article against the patient—site, syndrome, host, phenotype and timing—and returns strong match, partial match, mismatch or insufficient. Module B scores mNGS strength on an absolute axis (signal tier M1–M2 or read tier R3–R4) and a relative axis (dominance D2–D3, or rank 1 with read percentile ≥ 0.8). Module C grades same-organism non-mNGS evidence from C0 (none) to C3 (exact organism in a sterile-site or tissue specimen), and module D applies a specimen-colonization gate. Missing or pending tests are treated as unknown, never as negative. The final rule R5 (Fig. 1d) keeps every upstream pick and rescues a candidate only if C3 holds, or if A1 (at least two strong matches, or at least seven strong plus partial) coincides with site alignment and at least one B axis. Only then is a rationale written, by a language model constrained to the frozen selection and the patient’s evidence window, and audited for structure before delivery as a simple table and a traceable JSON, Markdown and XLSX package.

AETIA identifies causative pathogens more accurately than any single conventional read-out

We evaluated AETIA on [55] patients with suspected lower respiratory tract infection from Kaohsiung Medical University Hospital (KMUH, $n = [30]$), Taipei Veterans General Hospital (TVGH) and Tri-Service General Hospital (TSGH; together $n = [25]$), drawn from a three-hospital mNGS registry of 105 specimens (Table 1; Supplementary Fig. 1). Patients were predominantly critically ill (71% in intensive care at sampling), 56% had hospital- or ventilator-associated pneumonia, 37% had an active malignancy and 39% were receiving systemic corticosteroids; 60% of mNGS specimens were bronchoalveolar lavage fluid. The reference standard was the causative pathogen list adjudicated by the treating infectious-diseases physicians, with versioned label revisions (Methods). Because a patient may have more than one pathogen, we scored (patient, organism) pairs at exact species level with a frozen synonym table and report micro-averaged precision, recall and F1.

AETIA reached a precision of [71.6%], a recall of [91.4%] and an F1 of [80.3%] (Fig. 2a,b). The raw mNGS report, scored by taking every listed organism as a call, had the highest recall ([93.1%]) but a precision of [8.0%] (F1 [14.8%]): the report contains the pathogen almost every time, but for every causative organism it names it lists about [eleven] that are not $[(1-0.08)/0.08]$; confirm FP count from the evaluation JSON]. The targeted tests behaved in the opposite way. FilmArray and galactomannan together had a precision of [67.6%] but a recall of [39.7%] (F1 [50.0%]), and culture reached [48.1%] precision and [43.1%] recall (F1 [45.5%]). AETIA therefore kept essentially all of the sensitivity of mNGS while raising its precision by [ninefold], and it exceeded the precision of the targeted tests without their blind spots for fungi, viruses and fastidious organisms. [Add per-hospital metrics, exact-set-match rate and any-hit rate from the evaluation JSON; add 95% confidence intervals by patient-level bootstrap.]

AETIA reduces the organism burden presented to the physician

The clinical cost of mNGS is not only false attribution but the length of the list a physician must reconcile. In an earlier series of [35] intensive-care patients from KMUH evaluated with the first version of the pipeline, culture reported at least one organism in [24%] of patients and FilmArray in [64%], whereas mNGS reported at least one organism in every patient, with a mean of [5.76] organisms per patient (Fig. 2c,d). AETIA retained a call in every patient but presented a mean of [1.97] organisms, close to the [one to two] pathogens per patient in the adjudicated reference. The reduction is not a fixed cut-off: it comes from the demotion of colonizers, contaminants and reactivation viruses that fail the site, dominance and support rules, and from the guardrails that hold respiratory *Candida* and oral flora below the picking threshold unless independent evidence exists. [Confirm the relation of this 35-patient series to the 55-patient cohort and recompute organisms per patient for the 55 from the R5 decisions (518 candidates $\rightarrow$ 123 selected, i.e. 9.4 $\rightarrow$ 2.2 per patient).]

Structured integration outperforms direct prompting of frontier language models

To test whether the gain comes from the structure of the pipeline rather than from the language models inside it, we gave four frontier models (GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra) exactly the raw data that AETIA receives—the mNGS list with reads and control codes, cultures, FilmArray, galactomannan, molecular tests, imaging reports, blood counts, diagnosis and underlying conditions—for [41] patients (KMUH $n = [30]$, TVGH/TSGH $n = [11]$), de-identified, with a single expert-written prompt that explained the colonization and site rules in words, a JSON output schema and medium reasoning effort, and no retrieval, candidate pool or answer key (Methods; Supplementary Note [2]). Predictions were frozen before the reference labels were read.

All four models converged on the same operating point: precision between [67.7%] and [70.5%], recall between [56.6%] and [60.5%], and F1 between [62.4%] and [63.9%] (Fig. 3). On the same [41] patients, AETIA reached a recall of [90.8%] at a precision of [67.7%] ([69] true positives, [33] false positives, [7] false negatives), for an F1 of [77.5%], [13.6–15.1] points above the best direct-prompting model. The difference is almost entirely recall: prompted with the same warnings about colonization that AETIA encodes as rules, the models declined to call organisms that the rules, the hospital evidence and the case-matched literature together supported, whereas their precision was no better than the pipeline’s. [Slide 3 lists the pipeline’s precision on these 41 patients as 71.62%; the frozen tally TP 69 / FP 33 gives 67.65%, and the slide’s recall and F1 match that tally. Resolve from reports/idxfixd_20260906 before submission.] The newest model (GPT-6 Astra) improved recall by only [2.6] points over its predecessors, which suggests that the shortfall is not a matter of model capacity but of what a single prompt can be made to do with a long, noisy organism list.

Rescue by case-matched literature contributes a minority of selections

Across the [55] patients, the deterministic scorer considered [518] organism candidates and picked [114]; the R5 rule rescued a further [9], giving [123] selected organisms ([2.2] per patient). Rescue therefore changed [7%] of the final list, and every rescued organism carries its rule path (C3 direct, or A1 threshold with site alignment and B axis) and the PubMed identifiers of the articles that met the case-fit thresholds. [From the evaluation JSON: how many of the 9 rescued organisms were true positives; how many false negatives remained and why (not in report, below read tier, guardrail); how many false positives were upstream picks versus rescues.] The immutability of upstream picks means that rescue can only add recall; the precision of the final list is bounded by the deterministic stage.

Decisions are traceable to their evidence

Two cases illustrate the output (Supplementary Note [3]; details anonymized). In a patient with new bilateral consolidation and pleural effusions, a rising C-reactive protein and worsening hypoxaemia, the mNGS report listed *Klebsiella pneumoniae* with [68,569] reads alongside *Candida albicans*; AETIA selected *K. pneumoniae* (dominant, aligned, supported by imaging

and inflammatory markers) and excluded *C. albicans* as respiratory colonization with low read counts and repeated low-burden culture. In a second patient, *Candida tropicalis* was selected despite the general demotion of respiratory yeasts, because multiple blood-culture sets were positive for the same species and the bronchoalveolar mNGS reads were high, which satisfies the C3 direct-evidence path. [Both vignettes come from the January 2026 prototype (talk slide 15); regenerate them from the AETIA R5 rationale outputs for the same patients.] In each case the delivered record names the rule that fired, the modules that supported or opposed each organism, and the SHA-256 hash of every input file, so that a reviewer can reproduce the decision and see exactly which finding decided it.

The pipeline is reproducible offline

All [55] frozen R5 decisions were replayed from the frozen upstream, case-fit and clinical inputs and were byte-identical to the delivered files. The 41-patient direct-prompting comparison was likewise rescored from [164] frozen model outputs with identical metrics. The public code passes [319] Python and [8] Node offline tests that cover format processing, the deterministic scorer, the review, literature, case-fit and clinical rules, R5, rationale handoff, delivery and multi-cohort patient-identity isolation; the test harness blocks network access and strips API keys, so no test depends on a paid model or on live PubMed. The current scorer (v20) reproduces the picked set of every one of the [55] patients scored with the version used for the reported results (v19), while changing candidate levels in [15] and pick order in [5] patients (Supplementary Note [4]).

Discussion

mNGS has made the pathogen list longer, not the diagnosis clearer. The results here show that the interpretive step can be made explicit, reproducible and auditable without giving up the sensitivity that justifies ordering the test. On [55] patients from three hospitals, a pipeline of modality-specific extraction, deterministic scoring, language-model safety review and case-matched literature rescue recovered [91%] of physician-adjudicated pathogens at a precision of [72%], whereas the raw report, the multiplex panel and culture each delivered one of those two properties at the expense of the other. The same pipeline exceeded four frontier language models given identical data by [14–15] F1 points.

Why does structure beat prompting? The direct-prompting baselines were not naive: the prompt stated the colonization rules, the site rules and the meaning of the control codes, and the models reasoned at medium effort with a strict output schema. Their precision matched the pipeline’s. What they lost was recall, and the pattern of their misses is informative. Asked to be cautious about *Candida*, herpesviruses and environmental organisms, a model becomes cautious about everything, and declines the *Stenotrophomonas* with dominant reads in a ventilated patient or the *Pneumocystis* with modest reads in a patient on corticosteroids that the rules, the host tier and the literature together support [replace with real examples from the per-patient comparison]. A rule can encode the exception precisely—R2 reads suffice for a protected pathogen at rank 1, but not for a water-borne Gram-negative without dominance—whereas a prompt can only ask for judgement. The literature stage adds what neither a rule nor a general

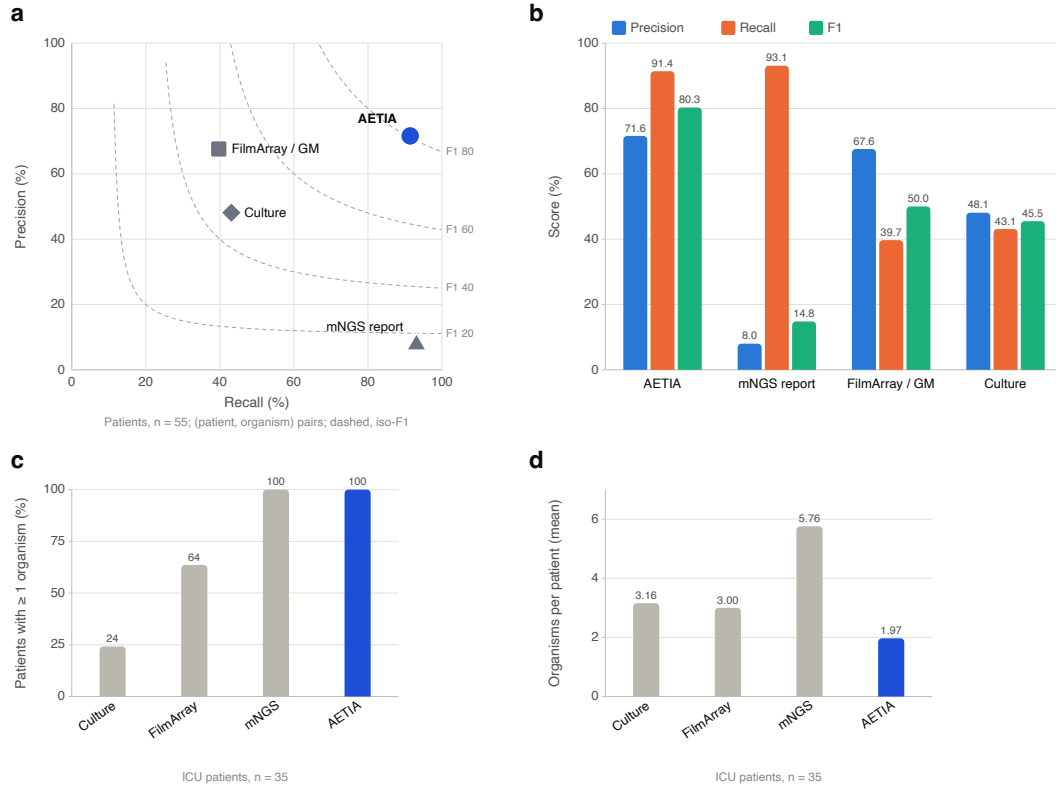


Figure 2: **Pathogen selection compared with the raw mNGS report, FilmArray/galactomannan and culture.** **a**, Precision against recall for AETIA (blue) and for each read-out taken at face value: all organisms in the mNGS report (triangle), all positive FilmArray or galactomannan targets (square) and all cultured organisms (diamond); dashed curves, constant F1. **b**, The same comparison as precision, recall and F1. $n = [55]$ patients (KMUH [30]; TVGH and TSGH [25]); unit, (patient, organism) pair at species level; reference standard, physician-adjudicated pathogens (Methods). **c,d**, Earlier intensive-care series (KMUH, $n = [35]$; first pipeline version): fraction of patients with at least one reported organism (**c**) and mean organisms per patient (**d**). [Values transcribed from slides; regenerate from the evaluation output.]

Table 1: **Characteristics of the three-hospital mNGS registry.** Values are n (%) or median (interquartile range). [Registry of 105 specimens from which the 55- and 41-patient cohorts were drawn (Supplementary Fig. 1); restrict to the evaluation cohorts once their manifests are available. Corticosteroid row reads both site encodings as yes; the workbook summary counts one (5.7%).]

Characteristic	All patients ($n = 105$)
<i>Hospital</i>	
KMUH	51 (48.6%)
TVGH	35 (33.3%)
TSGH	19 (18.1%)
Age, years	69 (56–75)
Male sex	68 (64.8%)
Current or former smoker	19 (18.1%)
Intensive care unit at sampling	75 (71.4%)
<i>Underlying conditions</i>	
Diabetes mellitus	29 (27.6%)
Chronic kidney disease	16 (15.2%)
End-stage renal disease on dialysis	4 (3.8%)
Liver cirrhosis	1 (1.0%)
Autoimmune disease	6 (5.7%)
Active malignancy	39 (37.1%)
Systemic corticosteroids [†]	40 (38.5%)
ARDS on the day of sampling [†]	48 (46.6%)
<i>Pneumonia category</i>	
Community-acquired	21 (20.0%)
Hospital- or ventilator-associated	59 (56.2%)
Not classified	25 (23.8%)
PaO ₂ /FiO ₂ ratio at sampling	200 (152–302)
<i>mNGS specimen</i>	
BALF	63 (60.0%)
Blood	24 (22.9%)
Sputum	4 (3.8%)
CSF	6 (5.7%)
Tissue	5 (4.8%)
Other	3 (2.9%)
Organisms per mNGS report, median (IQR)	4 (1–6)
mNGS report with no organism	8 (7.6%)
Any positive culture	69 (65.7%)
<i>Clinical course</i>	
Antimicrobial therapy changed	69 (65.7%)
Escalation	49 (46.7%)
De-escalation	19 (18.1%)
Change attributed to mNGS [‡]	52 (49.5%)
Mechanical ventilation within 3 days of sampling [†]	81 (82.7%)
In-hospital death [†]	51 (48.6%)

[†]Denominator excludes patients without a recorded value. [‡]Denominator is all patients; attribution was not recorded for 16 patients. KMUH, Kaohsiung Medical University Hospital; TVGH, Taipei Veterans General Hospital; TSGH, Tri-Service General Hospital; BALF, bronchoalveolar lavage fluid; CSF, cerebrospinal fluid.

prompt has: the actual evidence that this organism causes this syndrome at this site in this kind of host, judged article by article with the span that supports it. That the newest model (GPT-6 Astra) improved on its predecessors by only [2.6] recall points suggests the ceiling is set by the task formulation, not by model capacity.

The design choices that follow from auditability are worth stating as principles, because they apply beyond mNGS. First, language models extract and review; rules decide. Every organism that reaches the physician passed a rule whose inputs are recorded, so the decision can be replayed. Second, the answer key is quarantined: reference labels enter only the evaluation code, the case-fit prompt strips upstream picks, reads, cultures and review rationales, and the direct-prompting payloads were audited for label leakage before any model was called. Third, selection is frozen before explanation. Rationales are generated afterwards, restricted to the evidence window and checked for structure, so an eloquent paragraph cannot promote an organism the rules rejected. Fourth, upstream picks are immutable and rescue only appends. This bounds the precision of the final list by that of the deterministic stage and makes the contribution of each downstream module measurable: here, rescue changed [7%] of selections. These properties cost something—the pipeline calls a language model at four points and PubMed at one, and a full run on a new patient takes minutes and money rather than seconds—but they are what turns a research prototype into something a laboratory could validate.

Several limitations bound the claims. The study is retrospective and the cohort is small: [55] patients for the main comparison and [41] for the model comparison, from three hospitals in one country, sequenced by [one] laboratory. The reference standard is physician adjudication informed by all available results, including mNGS, so it is not independent of the test being evaluated, and its labels were revised in versioned steps during the study; we report the version used and the revision history (Methods), but a fully blinded reference would be stronger. The evaluation cohorts contain only patients with at least one adjudicated pathogen, so our precision estimates do not cover the false-positive risk in patients with no infection, which is where a pipeline built to preserve recall is most likely to err. The scoring rules and guardrails were developed by iterating on these same cases with the labels visible; the [55]-patient result is therefore a development-set result, not an untouched external test, and the gain over direct prompting should be read as a comparison of formulations on a shared cohort rather than as an estimate of prospective performance. The two vignettes and the organism-burden comparison come from an earlier [35]-patient series evaluated with the first version of the pipeline [confirm]. The literature and case-fit stages depend on a commercial language model and on live PubMed; frozen article snapshots make the reported decisions replayable, but a rerun on a new patient will retrieve different articles as the literature grows, and the case-fit judgements, although constrained by a traceability gate, are not deterministic. Finally, we measured agreement with adjudication, not clinical benefit: whether presenting two organisms instead of six changes antimicrobial choice, time to appropriate therapy or outcome remains to be shown.

These limitations define the next study. A prospective evaluation should freeze the rules and prompts before enrolment, include patients without infection, use an adjudication committee blinded to the pipeline output, and measure antimicrobial changes as an endpoint. Several extensions are straightforward within the present architecture. The reference-label revisions

and per-patient error analysis point to rule refinements that can be versioned and re-evaluated without touching the orchestrator. Antimicrobial-resistance genes on the same mNGS report, already parsed by the front end, could enter as a further evidence module. And because the mNGS signal features, the site logic and the literature stage are organism-agnostic, the pipeline can be extended to bloodstream and central nervous system infection, where the colonization problem is smaller but the stakes of a missed pathogen are higher. mNGS was meant to end the era of pneumonia treated without a diagnosis; a pipeline that can say which of the organisms it found is the diagnosis, and show why, brings that closer.

Methods

Ethics and study oversight

This retrospective study analysed existing clinical records and mNGS reports. Patients were not contacted, and no output of the pipeline was used in their care. The study was approved by the institutional review boards of Kaohsiung Medical University Hospital (KMUH; [protocol number]), Taipei Veterans General Hospital (TVGH; [protocol number]) and Tri-Service General Hospital (TSGH; [protocol number]), which [waived the requirement for informed consent because the data were collected retrospectively and analysed without direct identifiers / obtained written informed consent]. Direct identifiers (names, dates of birth, record and national identification numbers, addresses and contact details) were removed from every record before it was submitted to a commercial language-model service, and the reference labels were never part of any model input (below). [State the data-processing terms under which de-identified records were submitted to the OpenAI API (retention setting, hosting region, dates) and that the IRBs approved this external processing.] AETIA is research software: its output is a candidate list for physician review and is labelled as such on delivery.

Study design and participants

We conducted a retrospective, multicentre diagnostic-accuracy study. The index test was AETIA; the reference standard was the causative pathogen list adjudicated by the treating physicians; the comparators were the conventional microbiological read-outs taken at face value and four frontier language models prompted directly on the same data. Participants were patients who underwent mNGS for suspected lower respiratory tract infection at KMUH, TVGH and TSGH, three tertiary referral centres in Taiwan, between [month year] and [month year]. The hospitals maintain a shared registry of 105 mNGS specimens (KMUH 51, TVGH 35, TSGH 19) with demographics, comorbidities, conventional microbiology, mNGS reports, treatment changes and outcomes (Table 1). Two evaluation cohorts were fixed in cohort manifests, with per-patient input identity checks, before any model was queried (Supplementary Fig. 1). The R5 delivery cohort ($n = [55]$; KMUH [30], TVGH and TSGH [25]) comprised patients with a complete standardized record (nine sections, below), a laboratory-grouped mNGS report and at least one adjudicated causative pathogen. The direct-prompting cohort ($n = [41]$) comprised the [30] KMUH patients with at least one adjudicated pathogen and the [11] TVGH and TSGH patients whose records were complete in every section (record quality A). [Give the two-hospital

inclusion steps (54 registry candidates → 48 saved records → 25 delivered) with the reason for each exclusion; docs/Frontend.md records them as incompletely documented.] An earlier series of [35] KMH intensive-care patients, analysed with the first version of the pipeline, provided the organism-burden comparison (Fig. 2c,d) [state its overlap with the 55-patient cohort]. [Complete the STARD 2015 and TRIPOD+AI checklists and supply them as Supplementary Information.]

Metagenomic sequencing and conventional microbiology

mNGS was ordered by the treating physicians as part of clinical care; specimens were predominantly bronchoalveolar lavage fluid (60%) and blood (23%; Table 1). [Sequencing laboratory; nucleic-acid extraction and host depletion; library preparation; sequencing platform and run configuration; reads per sample; classification database and bioinformatic pipeline version; the laboratory’s reporting thresholds and the procedure that generates the control codes.] The laboratory report lists, per specimen, each reported organism with its read count and two control codes that record detection in the run’s no-template control (NTC) and in the [define the “RK” control; the abbreviation is not expanded anywhere in the source materials]; a code of 00 denotes absence from that control [confirm the code semantics (00, 8) with the laboratory]. Conventional microbiology—culture, the FilmArray multiplex PCR panel, galactomannan antigen testing and other molecular tests—was performed by each hospital’s clinical laboratory as part of routine care ([panel versions, galactomannan assay and cut-off index, culture media and specimen types]); results were taken from the registry and hospital records as reported, without reinterpretation.

Reference standard

The reference standard for each patient was the causative pathogen list (one to eight organisms) recorded in the registry by the treating infectious-diseases physicians at discharge, on the basis of all information available at that time, including the mNGS report, the clinical course and the response to treatment. [State who adjudicated (number and specialty of physicians per site), whether adjudication was independent or by consensus, and the criteria used to separate infection from colonization.] Labels were revised twice during the study, in versioned files: an answer-override file for the KMH cohort (2026-09-14) and a correction file for the direct-prompting cohort ([2026-09-12]), together affecting [four] patients; the label version used for every reported number is recorded with the evaluation output. Labels were read only by the evaluation scripts and were absent from every pipeline input, prompt and configuration, and the direct-prompting payloads were additionally screened for label-bearing fields before submission (below). The reference standard is not independent of the index test’s inputs, because the adjudicating physicians had seen the mNGS report, and it was not blinded to the pipeline’s output during rule development; both limitations are considered in the Discussion.

Data extraction and standardization

Source records were per-patient multi-sheet workbooks (KMUH) or a shared master workbook plus case-presentation slide decks (TVGH and TSGH). Workbooks were parsed deterministically into raw sections. Slide decks were parsed for text and embedded images, with an optional vision-model pass for image-only tables ([model and version used for the historical runs]), and linked to registry cases by patient, date and file cues. A language-model formatter (OpenAI gpt-5, temperature 1) converted each raw section into one of nine standardized JSON sections per patient—admission diagnosis, underlying conditions, complete blood count, other laboratory values, cultures, FilmArray, galactomannan, molecular microbiology and imaging—and the mNGS report formed a tenth input. Every record carries the time at which it was obtained. Records were restricted to a per-patient window anchored on the collection time of the mNGS specimen (the earliest specimen when several were sequenced); records without a parsable time were excluded rather than assumed contemporaneous. The mNGS report was ingested either as the laboratory’s grouped per-specimen list or as the flat list of organisms that passed the laboratory’s read-count and control filters; when both existed, the source was declared explicitly and never inferred. Because the formatter’s historical prompts were only partly recovered, the reported runs used the frozen formatter outputs; the current format processors were re-validated on [36] real-data runs and on the fully synthetic workbooks and decks that ship with the code. [The formatter model and temperature are the repository’s bindings (upstream/core/llm_parser.py); frontend/pipeline.py records them as user-supplied rather than as proof of the historical execution. Confirm from the original run logs.]

Patient identity was namespaced by hospital line. KMUH records were keyed by the hospital case code and joined to the mNGS specimen by specimen code; TVGH and TSGH records kept the registry’s global identifier, and the two-hospital line accepts only T- and V-prefixed cases. The two lines were staged in separate directories with SHA-256 hashes of every source file, and the pipeline refuses to merge cohorts whose identifiers could collide. All [297] KMUH and [480] two-hospital standardized sections used here were verified identical to the frozen inputs of the reported runs.

Overview of AETIA

AETIA processes one patient at a time through five stages in a fixed order (Fig. 1): modality agents read the standardized sections and a deterministic summary joins their outputs; a deterministic scorer grades every organism on the mNGS report and picks the causative candidates; a language-model safety review flags unpicked organisms for further evidence gathering without altering the picks; four evidence modules (literature, patient case-fit, sequencing strength and direct hospital evidence, plus a reported colonization gate) are computed for every carried candidate, and the final rule R5 fixes the delivered list; and a rationale is generated for the frozen selection, audited and exported. Language models are used only for extraction and judging, each with a fixed prompt and JSON output schema (Supplementary Table 5); every selection decision is made by explicit rules whose thresholds are held in versioned configuration files. Each stage writes its outputs together with the rule identifiers that fired and the SHA-256 hashes of its inputs, so that any decision can be traced to the finding that determined it. The model

identifiers and sampling settings given below are the repository’s configuration bindings at the time of writing ([confirm each against the run manifests of the reported runs, and give the dates on which each service was accessed]).

Modality agents and clinical summary

Each standardized section is read by a dedicated language-model agent (OpenAI gpt-5, temperature 1) with a fixed prompt and JSON output schema: a blood-count, laboratory and host agent that also assigns a host-vulnerability tier (V0–V3) and an opportunistic-coverage level (O0–O2) from immunosuppression, malignancy, corticosteroid exposure, renal failure and related conditions; a culture agent; a FilmArray and galactomannan agent; a molecular-microbiology agent; an imaging agent; and an mNGS-to-specimen agent that assigns each reported organism to a specimen and body site with a likelihood class (high, medium or low-colonizer). Each agent sees only its own section; the agents do not communicate, and none sees the reference labels. A deterministic summarizer (rule version v4) combines the agent outputs into per-organism support by modality (support, not support or unknown), specimen alignment (aligned; sterile or systemic; non-aligned) and a host context; a language-model summarizer used in earlier versions was retired. The host context sets a host-expansion flag when the vulnerability tier is V2–V3 or the opportunistic coverage is O1–O2; the flag relaxes the picking of level-3 organisms (below) but never promotes an organism by itself.

Features of mNGS organisms

Each organism on the mNGS report is described by five features (Supplementary Table 1). A control-subtracted priority from 1 (best) to 5 is assigned from the control codes, evaluated in order: 1 when NTC is 00 and RK is 00; 2 when RK is 8 and NTC is 00; 3 when RK is 00; 4 when RK is 8; and 5 when only NTC is 00. A code score (1.0, 0.85, 0.65, 0.45 and 0.30 for priorities 1–5) is combined with the percentile rank of $\log_{10}(\text{reads} + 1)$ among all organisms in the same run as $0.6 \times \text{code score} + 0.4 \times \text{percentile}$, and the result is labelled high (≥ 0.70), medium (≥ 0.45) or low. Read burden is tiered at decade thresholds: R0, 1–9 reads; R1, ≥ 10 ; R2, ≥ 100 ; R3, $\geq 1,000$; R4, $\geq 10,000$. Dominance is computed within each specimen record from the ratio of the top organism’s reads to the runner-up’s: D3 when ≥ 10 , D2 when ≥ 3 , D1 otherwise, and D0 for every organism that is not the top. When an organism appears in several specimen records of one patient, the record with the best priority and then the most reads is kept. Organism names are normalized through a canonical dictionary that merges synonyms (for example, CMV and human cytomegalovirus) but never merges species.

Deterministic scoring and picking

The scorer assigns each organism an mNGS signal tier from M1 (strong) to M5 (background) on the basis of its priority, read tier, percentile, dominance and likelihood class; the clauses are evaluated in order and the first satisfied tier applies (Supplementary Table 2). M1 requires priority 1–2 with R3–R4 reads, priority 1–2 with percentile ≥ 0.85 , or D3 dominance with at least R2 reads at priority ≤ 3 . M2 requires priority 1–2 with R2 reads and percentile ≥ 0.75 ; D2–D3

dominance with at least R2 reads at priority ≤ 3 ; same-organism support from a hospital test with at least R1 reads at priority ≤ 3 ; a protected pathogen (other than a reactivation virus or mould) with at least R1 reads, priority ≤ 3 and a non-low label; R3–R4 reads at priority ≤ 4 with a non-low label; *Pneumocystis jirovecii* with R1–R2 reads at priority ≤ 3 ; or a Mucorales organism at priority 1 with percentile ≥ 0.60 . M3 covers protected pathogens that meet a retention condition and, otherwise, priority 1–2 with R0–R2 reads, priority ≤ 3 with percentile ≥ 0.60 above trace reads, or priority 4 with R4 reads. M4 applies to priority 4–5, a low likelihood class, R0–R1 reads or a non-aligned specimen; M5 marks organisms on the background list. The protected-pathogen list comprises *P. jirovecii*, cytomegalovirus, herpes simplex virus types 1 and 2, *Toxoplasma gondii*, *Nocardia*, *Legionella*, *Mycobacterium tuberculosis*, *M. avium*, *M. kansasii*, *M. abscessus*, *Aspergillus*, the Mucorales, *Histoplasma* and *Cryptococcus*; reactivation herpesviruses and moulds have dedicated M2–M3 clauses that require non-host support or dominance.

Tiers map to causative levels 1–5. M1 becomes level 1 when the specimen is aligned or sterile and the organism is dominant or carries R3–R4 reads, and level 2 otherwise; M2 becomes level 2 when aligned and level 3 otherwise; M3 is level 3; M4 is level 3 only under host expansion when the organism remains plausible (not a reactivation virus, colonizer or non-aligned finding, and with non-host support, protected status, or priority ≤ 3 with R1 reads and a high or medium label), and level 4 otherwise; M5 is level 5. Before picking, 33 pre-specified guardrails (identifiers R-S4-*; Supplementary Table 3) demote organisms to level 4 or bar them from picking regardless of tier. They cover generic unidentified hospital labels; respiratory *Candida* without same-species invasive evidence, or supported only by non-invasive hospital tests; oral and aspiration flora, of which one representative is retained at level 3 when a compatible aspiration pattern exists; water-borne environmental Gram-negative bacteria without dominance or support; skin flora and coagulase-negative staphylococci; low-actionability viruses (Epstein–Barr virus, human herpesviruses 6 and 7) and non-core respiratory viruses; weak mould signals with host-only support; mNGS-only R0–R1 non-protected organisms; common hospital organisms with low, non-dominant reads; non-pulmonary culture context; retention of one organism per taxonomic group; and a rule that prevents trace-read protected pathogens from being picked without support.

Organisms at level 1–2 are picked. Level-3 organisms are picked only with a protected or supported rationale: protected-pathogen status; an oral-flora or respiratory-commensal representative rule; a compatible typical-pneumonia context; direct support from culture, FilmArray, galactomannan or molecular testing; or, under host expansion, priority ≤ 2 with at least R2 reads or D2–D3 dominance. *Candida* and generic yeasts are never picked at level 3 without invasive support. When nothing is pickable, the best level-4 organisms that pass the exclusion rules are reported as “best available” and flagged as such. Picked organisms are ordered by level, then signal tier, then reads, and the first level 1–2 organism is labelled primary. Every organism, picked or not, is written out with its level, the rule identifiers that fired and an exclusion reason code. The reported decisions were produced with scorer version v19 (mngs_deterministic_level_only_v19_low_specificity_taxonomy); the current version v20 (v20_central_taxonomy_profile) centralizes the taxonomy profiles and reproduces the picked

set in all [55] patients, with candidate-level differences in [15] and pick-order differences in [5] (Supplementary Note 4).

Safety review of unpicked organisms

Organisms that were not picked are queued for review by a language model (OpenAI gpt-5) that receives the deterministic output, the hospital-side summary and the mNGS-to-specimen classes. The prompt (Supplementary Note 1) instructs the model that it may not replace or reorder the picked list, that immunocompromise and protected status are not by themselves grounds for upgrade, and it restates the colonization, reactivation, oral-flora, environmental and *Candida* rules with explicit read, rank and dominance conditions. The model assigns each organism to one of four tiers: high priority and context needed, which are carried into the evidence stage, and low specificity and omitted with reason, which are retained for audit only. The merge step appends the carried organisms to the deterministic output without altering it and records the tier of every organism. In the replay of the [55] reported patients, picked sets and tier membership were identical in all patients.

Literature evidence (module A)

For each carried candidate, and for each upstream pick (literature on picks was enabled in the reported runs), PubMed was queried through the NCBI E-utilities with the organism name in the title or abstract, combined with a disjunction of lower respiratory tract infection site terms and restricted to publications up to 30 June 2026; at most ten articles were retrieved, and every query and response was cached with its hash. Each title and abstract was judged by a language model (OpenAI gpt-5.6-luna, reasoning effort low, at most 1,200 output tokens, prompt version `article_support_v1`) for whether it provides human clinical evidence that the exact species causes infection at the target site. The judge returns a verdict (support, against, irrelevant or unclear), the organism scope (exact species, genus only or other), human-clinical and target-site indicators, and an evidence span that must be a contiguous excerpt of at most 20 words from the supplied text. A support verdict counts only when the scope is the exact species, the evidence is human and clinical, the site matches and the span is valid. Module A is positive when at least three valid supports exist among at least five judgeable articles; with fewer than five judgeable articles, or when retrieval fails, it abstains rather than voting negative. A literature-off branch exists for cost control and was not used for any reported number.

Patient case-fit (module A1)

Each retrieved article is re-judged against a case card built from the patient's standardized record. The case card contains the specimen and site, the clinical syndrome, host factors, phenotype (imaging and laboratory patterns) and timing; it deliberately excludes the reference labels, the upstream picks, read counts, culture and PCR results and any review rationale, so that the judgement concerns the fit between the published patients and this patient rather than the strength of this patient's evidence. The model (OpenAI gpt-5.6-luna, reasoning effort low, at most 1,400 output tokens, prompt version `article_casefit_v1`) extracts the organism

scope, the human clinical context (causal infection; colonization or contamination; reactivation or bystander; detection only; non-human; unclear), site and syndrome alignment, and four patient-specific dimensions (specimen, host, phenotype and temporal alignment), each rated match, partial, mismatch or unknown. A deterministic rule, not the model’s own conclusion, derives the verdict: mismatch when the scope is not the exact species, the evidence is non-human, or the site or syndrome conflicts; insufficient when the hard gates (exact species, causal infection, and at least partial site and syndrome alignment) are not all met; strong match when site and syndrome match, at least one patient-specific dimension matches and none conflicts; and partial match otherwise. Colonization at a compatible site counts as counter-evidence. A traceability gate requires verbatim supporting and counter-evidence spans before an article enters aggregation. The numbers of strong and partial matches per candidate are the A1 signals used by R5. When frozen articles were missing for a candidate, the batch fell back to live PubMed retrieval; [15] of [62] case-fit artifacts record such a fallback, and the reported decisions replay from the frozen artifacts.

Sequencing strength, direct evidence and colonization gate (modules B, C and D)

Module B evaluates mNGS strength on two axes from the frozen candidate features. The absolute axis is positive when the signal tier is M1–M2 or the read tier is R3–R4; the relative axis is positive when dominance is D2–D3 or the priority is 1 with read percentile ≥ 0.8 . Site alignment is taken from structured candidate-specific evidence when present and from the upstream alignment label otherwise; an explicit site mismatch cannot be overridden by the generic label. R5 requires either axis together with site alignment; the stricter conjunction of both axes is computed and reported but not used as a gate. Module C grades same-organism evidence from non-mNGS tests, using only structured records with a usable specimen site and an exact organism match: C3 when a high-specificity assay (culture, a molecular test or FilmArray with an explicit organism) is positive in a sterile-site or tissue specimen; C2 when such an assay is positive at the target site; C1 when only context is available (a positive record without a usable site, a record flagged as contaminated, a modality-level support summary, or respiratory *Candida* supported only by culture); CNEG when an exact-organism structured test at the target site is explicitly negative; and C0 when no direct evidence exists, which includes pending, ordered and not-performed tests. Module D checks site coherence and colonization guardrails and returns pass, guarded, block or unknown; it is reported with every decision but is not a gate in R5. A temporal-coherence module (three days before and after the anchor) is computed for research arms only. The clinical configuration is versioned (`clinical_v1`) and marked exploratory; changing any threshold requires a new configuration version (Supplementary Table 4).

Final selection rule (R5)

For every candidate carried into the evidence stage, R5 computes

$$\begin{aligned} \text{selected} = & \text{picked} \vee \text{C3} \vee [(A1_{\text{strong}} \geq 2 \vee A1_{\text{strong}} + A1_{\text{partial}} \geq 7) \\ & \wedge \text{site aligned} \wedge (B_{\text{absolute}} \vee B_{\text{relative}})], \end{aligned}$$

where picked denotes an upstream pick and the remaining terms are the module outputs defined above (rule identifier R5_A1_UNION_AND_B_LOOSE; thresholds in Supplementary Table 4). Upstream picks are selected unconditionally and keep their order; rescued organisms are appended in alphabetical order, which carries no clinical ranking. For every patient, the rule path of each selected organism (direct evidence or case-fit with site alignment and sequencing strength), the gate values, the disposition of every non-selected candidate, the source file hashes and a manifest are written, together with the statement that failing the rescue rule does not exclude infection. The earlier pruning arms (a stricter B, D as a gate) are retained as research comparisons and were not used for any reported list.

Rationale generation and delivery

After R5, the patient's structured record is cut to an evidence window of two calendar days on either side of the mNGS anchor for candidate-specific microbiology ([confirm the window from the run manifest; two days is the template default]), with the full time course retained in a separate view, and a rationale is generated for each selected organism by a language model (OpenAI gpt-5.6-luna, reasoning effort low) from these views and the frozen decision. The generator is instructed that selection and ranking are fixed. Its output is audited for structure (one entry per selected organism, no organism outside the selection, all required fields present), and any failure blocks delivery. A first-pass rationale written against the upstream picks is audited and then superseded by a second pass aligned to the R5 selection, so that no delivered explanation refers to a superseded list. Delivery exports a simple table (hospital, patient, selected pathogens and reasons) and a traceable JSON, Markdown and XLSX package in which every organism links to its rule path, module results, article identifiers and input hashes, followed by an automated delivery audit. Delivered outputs are labelled as research results pending clinical review.

Conventional read-out comparators

Each conventional read-out was scored at face value by treating its reported organisms as that read-out's predictions: every organism on the mNGS report; every FilmArray or galactomannan target reported positive; and every organism isolated in culture within the evidence window. [Confirm these definitions and the window from the baseline script; state how mixed or normal flora, "no growth" and pending results were handled, and whether galactomannan positivity was scored as *Aspergillus* species.] No read-out was reinterpreted or filtered, so these comparators represent the information a physician receives before any integration.

Direct-prompting comparison

To test whether the pipeline’s advantage derives from its structure rather than from the language models it contains, four frontier models were given, for each of the [41] patients, a single request containing the nine standardized clinical sections and the raw mNGS list (organism, reads, control codes and specimen), after removal of direct identifiers and of every field that could carry an answer (reference labels, upstream picks, review tiers and rationales). The mapping from patient to payload was frozen and verified by identity checks. After a correction to that mapping, 22 responses per model (66 of the 123 responses of GPT-5.6 Sol, GPT-5.6 Luna and GPT-5.6 Terra) were regenerated from the corrected payloads and the remaining 57 were verified byte-identical to their originals, so that all 123 responses of the three earlier models and all 41 responses of GPT-6 Astra correspond to the corrected payloads. The prompt (Supplementary Note 2; written in Traditional Chinese by the study team) casts the model as an infectious-diseases reviewer, states that detection is not causation, lists the colonization, site, *Candida*, environmental and reactivation cautions, allows zero or more pathogens, requires species-level names and demands a JSON object with predicted pathogens, supporting evidence, counter-evidence, excluded organisms and limitations. The models were OpenAI `gpt-5.6-sol`, `gpt-5.6-luna`, `gpt-5.6-terra` and `gpt-6-astra` ([dates of access]), called through the Responses API with reasoning effort medium, a JSON schema and at most 4,000 output tokens, without tools, retrieval or conversation history; each model was run once per patient. Every response was stored with the prompt hash, model identifier, reasoning setting and payload hash, and validated against them before scoring.

Outcome measures and scoring

The unit of evaluation was the (patient, organism) pair. Predicted and reference organism names were normalized (Unicode NFKC, case and whitespace) and mapped through a frozen synonym table (for example, CMV to human cytomegalovirus, PJP to *P. jirovecii* and CRKP to *Klebsiella pneumoniae*) plus per-cohort aliases. Matching was at exact species level; genus-level relaxation was not applied, so a prediction of *Candida* against a reference of *Candida tropicalis* counts as one false positive and one false negative. The primary outcome measures were precision, recall and F1 micro-averaged over pairs. Secondary measures, computed by the same script, were the per-patient (macro-averaged) values, the proportions of patients with at least one reference organism identified, with all reference organisms identified and with an exact set match, and the number of patients for whom a method returned no organism (Supplementary Note 5). Because both evaluation cohorts were restricted to patients with at least one adjudicated pathogen, the reported precision does not capture false positives in patients without a pathogen.

Computational reproducibility

The pipeline is orchestrated by a single entry point that validates the cohort manifests and per-stage inputs, runs the stage programs in a fixed order as subprocesses and records every command, working directory and output hash in a run manifest; the orchestrator contains no selection logic, and a plan mode lists all stages and inputs without side effects. Output

directories are never overwritten. The code carries SHA-256 manifests of [382] research source files and [470] package files, and editing any listed file fails source verification. The offline test suite ([319] Python and [8] Node tests) runs under a site customization that disables network sockets and removes API keys from the environment, so no test depends on a paid model or on live PubMed. The [55] reported R5 decisions, the [62] case-fit artifacts and the [164] direct-prompting outputs were replayed from their frozen inputs and reproduced the delivered files or metrics exactly; the v19 scorer used for the reported runs was recovered and verified against all [55] saved scorer outputs. The reported runs used Python 3.13 and Node 24 on Windows ([OpenAI SDK version; hardware]); macOS and Linux are untested.

Statistics and reproducibility

No statistical method was used to predetermine sample size; the cohorts comprise every registry patient who met the inclusion criteria when the cohort manifests were frozen, and the small number of eligible patients is a limitation (Discussion). Patients were not randomized, because every method was applied to the same patients. Blinding was not applicable to the deterministic stages, which are exact functions of their inputs; the language-model stages received no reference labels, and predictions of every method were frozen before the reference labels were read by the evaluation scripts. The investigators who wrote the scoring rules and guardrails were not blinded to the reference labels: the rules were developed by iterating on the same cases with the labels visible, and no patient was held out from rule development, so the reported performance is a development-set result (Discussion) [confirm that no case was held out]. Patients were excluded only for missing inputs, before any model was queried (Supplementary Fig. 1); no patient or prediction was excluded afterwards. The clinical records have never been made public, so the evaluation cohort cannot have been part of the training data of the language models. Precision, recall and F1 are point estimates from single runs: the deterministic stages replay identically, whereas the language-model stages were run once per patient (formatter, agents and safety review at temperature 1; article, case-fit and rationale generation at reasoning effort low; direct prompting at reasoning effort medium) and their outputs frozen, so run-to-run variability of these stages is not captured by the reported numbers. [Add before submission: 95% confidence intervals for every precision, recall and F1 by patient-level bootstrap (2,000 resamples of patients with replacement, pairs pooled within each resample); paired bootstrap differences in F1 between AETIA and each comparator on the same patients; an exact McNemar test on per-pair correctness where the pair sets coincide; and, if the budget allows, two further direct-prompting runs per model to report the run-to-run range. Report n as patients and as pairs in every legend.]

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article. [Complete the Reporting Summary; its statistics, software and data sections must match this Methods section (single-run point estimates, bootstrap CIs once added, Python 3.13 and Node 24, and the availability statements below).]

Data availability

The patient-level clinical records, mNGS reports, reference labels, model inputs and model outputs contain protected health information and cannot be shared publicly. Aggregate cohort characteristics are given in Table 1. De-identified per-patient evaluation tables (predicted and reference organism sets for AETIA, each conventional read-out and each directly prompted model, with the label version) will be [\[provided as Supplementary Data / made available from the corresponding author on reasonable request, subject to approval by the three institutional review boards and a data-use agreement; requests will be answered within four weeks\]](#). The PubMed identifiers and frozen judgements of every article used by modules A and A1 will be [\[provided as Supplementary Data\]](#). Source data for Figs. 2 and 3 are provided with this paper.

Code availability

The AETIA pipeline, including all rules, prompts, configuration files, synthetic example data and the offline test suite, is maintained in the `Multimodal-Diagnosis-Model` repository ([\[URL; the repository is private pending co-author and institutional licence approval, LICENSE_STATUS.md\]](#)) and will be released under [\[licence\]](#) with the version tag and commit used for the reported runs archived at Zenodo ([\[DOI\]](#)). The scripts that generate the figures and tables of this manuscript are included with the manuscript source.

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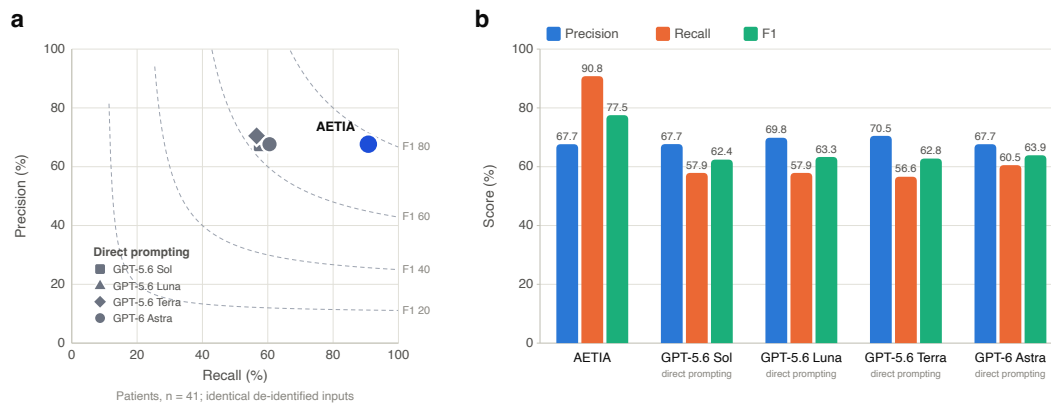


Figure 3: Comparison with frontier language models prompted directly on the same data. GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra received the identical de-identified record—raw mNGS list with reads and control codes, cultures, FilmArray, galactomannan, molecular tests, imaging reports, blood counts and diagnoses—with one prompt, a JSON schema and no retrieval or reference labels (Methods; Supplementary Note [2]). **a**, Precision against recall; AETIA in blue, models in grey (marker key in panel); dashed curves, constant F1. **b**, Precision, recall and F1. $n = [41]$ patients (KMUH [30]; TVGH and TSGH [11]); micro-averaged over (patient, organism) pairs; AETIA values from the frozen R5 decisions on the same patients (TP [69], FP [33], FN [7]). [Slide gives 71.62% precision; TP/FP imply 67.65%.]